

Structure and organization of prokaryotic and eukaryotic cells; Structure and function of cell organelles; cell cycle; Bio-membranes; Cytoskeletal elements; Chromosome structure; Karyo type; Chromatin organization

Fundamental Processes: Photosynthesis; Cellular respiration; Movement through cell membrane; Nutrition; Blood clotting; Human physiological systems; Replication; Transcription; Translation; DNA repair mechanisms; Plant physiology; Bacterial growth; Microbial genetics, Secondary metabolites

Cell Communication and Cell Signaling: Tight adherents and communicating cell junctions; Cell adhesion molecules; Cadherins and Integrins; Extracellular matrix; Cell cycle; Basics of cancer; Basics of cell signaling; Major signaling pathways.

Developmental Biology: Stages of development; Mechanism of differentiation; Germ layers; Potency; Morphogenetic movements; Early and late development in model organisms; Cell division; Gametogenesis and fertilization in animals and flowering plants; Embryology; Seed germination; Dormancy; Evolution and natural selection; Mendel's law of heredity; Evidence of DNA as genetic information carrier; Hardy-Weinberg law; Extra-chromosomal inheritance; Sex-linked inheritance in humans; Mutations

Plant and Animal Biotechnology: Plant tissue culture techniques; Totipotency; Organogenesis and Somatic embryo genesis; Suspension culture; Protoplast isolation and somatic hybridization; Production of secondary metabolites; Basic techniques in animal cell and organ culture; Bioreactors for large scale culture of animal cells; Stem cells; Transgenic plants and animals

Immunology and Immunotechnology: Immunity; Antigen; Structure of antibody; Hapten; Antigen-antibody interaction, Introduction to antigen presentation; Role of MHC; Complement system; Bacterial diseases of humans; Types of vaccines; Immunization; Recombinant vaccines.

Inheritance Biology: Mendelian's principles,

Extensions of Mendelian principles, Gene mapping methods, Extra chromosomal inheritance, Human genetics, Mutations, Structural and numerical alterations of chromosomes.

Diversity of Life Forms: General characteristics of life forms; General characteristics of bacteria, fungi, algae, Microbial growth curve; plant and animal viruses; Classification of plant and animal kingdom

Ecological Principles and Environmental Biology: Ecosystem; Ecological relationships; Habitat and niche; Ecology of eco systems; Air, water, and soil pollution; Greenhouse effect and global warming; Noise pollution; Pollution abatement; Wastewater treatment; Disposal of solid wastes; Bio geochemical cycles of elements; Bio remediation; Bio leaching; Bio pesticides; Bio fertilizers

Evolution and Behaviour: Evolution and natural selection; Mendel's law of heredity; Evidences of DNA as genetic information carrier; Hardy-Weinberg law; Extra-chromosomal inheritance; Sex-linked inheritance in humans; Mutations.

Applied Biology: Basics of fermentation technology; Microbes in industry; Biosensors; Biofuels; Principles of gene cloning; Methods of gene transfer; Application of biology in agriculture, health, industry, and environment sectors.

Assay techniques in Biology: Basics of Centrifugation; Electrophoresis; Chromatography; Microscopy; UV- Visible spectrophotometry; Radiotracer technique; PCR; DNA sequencing; Southern blotting; Tests of significance; Analysis of variation; Correlation and regression; Hybridoma technology; Basic techniques in bio informatics

Bio informatics and Computational Biology: Sequence and Structural Databases (NCBI, GenBank, EMBL, DDBJ, PDB); SNP databases; Visualization tools- Pymol, VMD; Functional Annotation; Local and Global Alignment; Phylogenetics; Pharmacogenomics; Machine learning

Diagnostic techniques: X-Rays, CT scan, MRI, Pathology test, ECG, EEG.

18. Centre for Community Development and Research (CCDR)

Discipline: Community Development and Research

- **Theories of Development:**

Sustainable development, rural development models, human development approach, and community empowerment.

Principles and skills: Leadership development, building partnerships, and involving NGOs/CBOs.

Rural and Urban development: Infrastructure, migration, and slum redevelopment.

- **Social & Economic Context:**

Poverty, inequality, and human development.

Gender dynamics, social exclusion, and marginalized communities.

Agrarian change, rural economy, and self-help groups (SHGs).

- **Policy & Governance:**

Indian planning process and constitutional provisions (Directive Principles).

Decentralization: Panchayati Raj institutions and local governance.

Sustainable development goals (SDGs) and social policy formulation.

Make in India policy. World strategic issues.

- **Environmental & Emerging Issues:**

Climate change impact, disaster management, and sustainable technologies.

Corporate Social Responsibility (CSR) and its role in development.

Quantitative/Qualitative Techniques: Basic social statistics, development indices, and qualitative data analysis techniques.

- **Research Fundamentals**

Types of research (qualitative, quantitative, applied, analytical).

Research design, hypothesis formulation, and literature review.

Data collection methods (survey, interview, case study).

Sampling techniques (probability & non-probability).

Statistical tools: mean, median, mode, ANOVA, t-test, chi-square, and correlation/regression.

Research ethics, plagiarism, report writing, referencing.

Project proposal preparation.

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and
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